

Exercise J01

Q1. If $\frac{T_2}{T_3}$ in the expansion of $(a + b)^n$ and $\frac{T_3}{T_4}$ in the expansion of $(a + b)^{n+3}$ are equal, then n is equal to

- (A) 3 (B) 4 (C) 5 (D) 6

Ans. C

Sol. -

Q2. If number of terms in the expansion of $(x - 2y + 3z)^n$ are 45, then n is equal to

- (A) 7 (B) 8 (C) 9 (D) None of these

Ans. B

Sol. -

Q3. The greatest coefficient in the expansion of $(1 + x)^{2n+2}$ is

- (A) $\frac{(2n)!}{(n)!}$ (B) $\frac{(2n+2)!}{[(n+1)!]^2}$ (C) $\frac{(2n+2)!}{n!(n+1)!}$ (D) $\frac{(2n)!}{n!(n+1)!}$

Ans. B

Sol. -

Q4. The expression $\frac{1}{\sqrt{4x+1}} \left[\left[\frac{1+\sqrt{4x+1}}{2} \right]^7 - \left[\frac{1-\sqrt{4x+1}}{2} \right]^7 \right]$ is a polynomial in x of degree

- (A) 7 (B) 5 (C) 4 (D) 3

Ans. D

Sol. Given expression can be rewritten as

$$\frac{2}{2^7 \sqrt{4x+1}} \left[{}^7C_1(\sqrt{4x+1}) + {}^7C_3(\sqrt{4x+1})^3 + \dots + {}^7C_7(\sqrt{4x+1})^7 \right]$$

$$\frac{1}{2^6} \left[{}^7C_1 + {}^7C_3(\sqrt{4x+1})^2 + \dots + {}^7C_7(\sqrt{4x+1})^6 \right]$$

$\therefore$ Last term becomes $(4x+1)^3$

Hence degree is 3

Q5. If the coefficient of $(r + 2)^{\text{th}}$ and $(3r)^{\text{th}}$ term in the exp. of $(1 + x)^{2n}$ are equal then

- (A) $n = 2r + 1$ (B) $n = 2r - 1$ (C) $n = 2r$ (D) None of these

Ans. C

Sol. -

Q6. The coefficient of the middle term in the binomial expansion in powers of x of $(1 + \alpha x)^4$ and of $(1 - \alpha x)^6$ is the same if α equals-

- (A) $-\frac{5}{3}$ (B) $\frac{10}{3}$ (C) $-\frac{3}{10}$ (D) $\frac{3}{5}$

Ans. C

Sol. Coefficient of middle term in $(1 + \alpha x)^4 = {}^4C_2 \alpha^2$
 coefficient of middle term in $(1 - \alpha x)^6 = {}^6C_3 (-\alpha)^3$

$${}^4C_2 \alpha^2 = - {}^6C_3 \alpha^3$$

$$-\frac{6}{20} = \alpha$$

$$\alpha = \frac{3}{10}$$

Q7. If the coefficients of r^{th} , $(r + 1)^{\text{th}}$ and $(r + 2)^{\text{th}}$ terms in the binomial expansion of $(1 + y)^m$ are in A.P., then m and r satisfy the equation -

- (A) $m^2 - m(4r - 1) + 4r^2 - 2 = 0$ (B) $m^2 - m(4r + 1) + 4r^2 + 2 = 0$
 (C) $m^2 - m(4r + 1) + 4r^2 - 2 = 0$ (D) $m^2 - m(4r - 1) + 4r^2 + 2 = 0$

Ans. C

Sol. Coefficient of r^{th} , $(r + 1)^{\text{th}}$ and $(r + 2)^{\text{th}}$ terms in expansion of $(1 + x)^n$ are ${}^nC_{r-1}$, nC_r , ${}^nC_{r+1}$

$$\text{Then } {}^{2n}C_r = {}^nC_{r-1} + {}^nC_{r+1}$$

$$\Rightarrow n^2 - n(4r + 1) + 4r^2 - 2 = 0$$

Short Method: Let $r = 1$, hence nC_0 , nC_1 and nC_2 are in A.P.

$$\Rightarrow 2 \cdot {}^nC_1 = {}^nC_0 + {}^nC_2 \Rightarrow 2n = 1 + \frac{n(n-1)}{2}$$

$$\Rightarrow 4n = 2 + n^2 - n \Rightarrow n^2 - 5n + 2 = 0$$

Which is given by (b)

Q8. If the coefficient of x^7 in $\left[ax^2 + \left(\frac{1}{bx}\right)\right]^{11}$ equals the coefficient of x^{-7} in $\left[ax - \left(\frac{1}{bx^2}\right)\right]^{11}$, then a and b satisfy the relation -

- (A) $a - b = 1$ (B) $a + b = 1$ (C) $\frac{a}{b} = 1$ (D) $ab = 1$

Ans. D

Sol. In the expansion of $\left(ax^2 + \frac{1}{bx}\right)^{11}$, the general term is

$$T_{r+1} = {}^{11}C_r (ax^2)^{11-r} \left(\frac{1}{bx}\right)^r = {}^{11}C_r a^{11-r} \frac{1}{b^r} x^{22-3r}$$

For x^7 , we must have $22 - 3r = 7 \Rightarrow r = 5$, and the coefficient of $x^7 = {}^{11}C_5 \cdot a^{11-5} \frac{1}{b^5} = {}^{11}C_5 \frac{a^6}{b^5}$

Similarly, in the expansion of $\left[ax - \left(\frac{1}{bx^2}\right)\right]^{11}$, the general term is $T_{r+1} = {}^{11}C_r (-1)^r \frac{a^{11-r}}{b^r} x^{11-3r}$

For x^{-7} we must have $11 - 3r = -7 \Rightarrow r = 6$, and the coefficient of x^{-7} is ${}^{11}C_6 \frac{a^5}{b^6} = {}^{11}C_5 \frac{a^5}{b^6}$

As given, ${}^{11}C_5 \frac{a^6}{b^5} = {}^{11}C_5 \frac{a^5}{b^6} \Rightarrow ab = 1$

Q9. Let x be the 7th term from the beginning and y be the 7th term from the end in the expansion of $\left(3^{1/3} + \frac{1}{4^{1/3}}\right)^n$. If $y = 12x$ then the value of n is :

- (A) 9 (B) 8 (C) 10 (D) 11

Ans. A

Sol. -

Q10. In the binomial expansion of $(a - b)^n$, $n \geq 5$, the sum of 5th and 6th terms is zero, then $\frac{a}{b}$ equals-

- (A) $\frac{5}{n-4}$ (B) $\frac{6}{n-5}$ (C) $\frac{n-5}{6}$ (D) $\frac{n-4}{5}$

Ans. D

Sol. $(a-b)^n \Rightarrow$ general term $T_{r+1} = {}^nC_r \cdot a^{n-r} \cdot b^r (-1)^r$

$${}^nC_4 \cdot a^{n-4} \cdot b^4 - {}^nC_5 \cdot a^{n-5} \cdot b^5 = 0$$

$${}^nC_4 \left(\frac{a}{b}\right) = {}^nC_5 \Rightarrow \frac{n!}{4!(n-4)!} \left(\frac{a}{b}\right) = \frac{(n-4)}{5}$$

$$\left(\frac{a}{b}\right) = \frac{(n-4)}{5}$$

Q11. If $(1 + ax)^n = 1 + 8x + 24x^2 + \dots$ then the value of a and n is

- (A) 2, 4 (B) 2, 3 (C) 3, 6 (D) 1, 2

Ans. A

Sol. -

Q12. The coefficient of x^4 in the expansion of $(1 + x + x^2 + x^3)^n$ is

- (A) nC_4 (B) ${}^nC_4 + {}^nC_2$ (C) ${}^nC_4 + {}^nC_2 + {}^nC_4 \cdot {}^nC_2$ (D) ${}^nC_4 + {}^nC_2 + {}^nC_1 \cdot {}^nC_2$

Ans. D

Sol. -

Q13. The co-efficient of x^9 in the polynomial given by ,

$(x + 1)(x + 2) \dots (x + 10) + (x + 2)(x + 3) \dots (x + 11) + \dots + (x + 11)(x + 12) \dots (x + 20)$ is :

- (A) 5511 (B) 5151 (C) 1515 (D) 1155

Ans. D

Sol. Coefficient of x^9 in $(x + 1)(x + 2) \dots (x + 10) + (x + 2) \dots (x + 11) + \dots + (x + 11)(x + 12) \dots (x + 20)$

Let $P(x) = (x + 1)(x + 2) \dots (x + 10)$

$\Rightarrow x = -1, \dots, -10$ are roots of $P(x)$

Coefficient of $x^9 = -(-1 - 2 \dots -10) = 1 + \dots + 10$

$$= \frac{10 \times 11}{2} = 55$$

Coefficient of x^9

$= 55 + 65 + 75 + 85 + 95 + 105 + 115 + 125 + 135 + 145 + 155 \rightarrow$ A.P. with $d = 10$

$$= \frac{11}{2} [110 + 10 \times 10] = \frac{210}{2} \times 11$$

$$= 105 \times 11 = 1155$$

Q14. The coefficient of x^{39} in the expansion of $\left(x^4 - \frac{1}{x^3}\right)^{15}$ is-

- (A) - 455 (B) - 105 (C) + 455 (D) + 105

Ans. A

Sol. -

Q15. The coefficient of x^n in expansion of $(1 + x)(1 - x)^n$ is-

- (A) $(n - 1)$ (B) $(-1)^n (1 - n)$ (C) $(-1)^{n-1} (n - 1)^2$ (D) $(-1)^{n-1} n$

Ans. C

Sol. Coefficient of x^n in expansion of $(1 + x)(1 - x)^n$ i.e., coefficient of x^n in expansion of $(1 - x)^n +$ coefficient of x^{n-1} in expansion of $(1 - x)^n$]

Now, $(-1)^n {}^nC_n + (-1)^{n-1} {}^nC_{n-1}$
 $(-1)^n [{}^nC_n - {}^nC_{n-1}] = (-1)^n [1 - n].$

Q16. For natural numbers m, n if $(1 - y)^m (1 + y)^n = 1 + a_1 y + a_2 y^2 + \dots$, and $a_1 = a_2 = 10$, then (m, n) is—

- (A) (35, 20) (B) (45, 35) (C) (35, 45) (D) (20, 45)

Ans. C

Sol. $(1 - y)^m (1 + y)^n = 1 + a_1 y + a_2 y^2 + \dots$
 $(1 - my + {}^mC_2 y^2 \dots) (1 + ny + {}^nC_2 y^2 \dots) = 1 + a_1 y + a_2 y^2 + \dots$
 $a_1 = n - m = 10 \dots (1)$
 $a_2 = mC_2 + nC_2 - mn = 10 \dots (2)$
solving (1) & (2), we get $(m, n) \equiv (35, 45)$

Q17. The coefficient of x^7 in the expansion of $(1 - x - x^2 + x^3)^6$ is :

- (A) -144 (B) 132 (C) 144 (D) -132

Ans. A

Sol. -

Q18. If the coefficient of x^3 and x^4 in the expansion of $(1 + ax + bx^2) (1 - 2x)^{18}$ in powers of x are both zero then (a, b) is equal to :

- (A) $\left(14, \frac{272}{3}\right)$ (B) $\left(16, \frac{272}{3}\right)$ (C) $\left(16, \frac{251}{3}\right)$ (D) $\left(14, \frac{251}{3}\right)$

Ans. B

Sol. -

Q19. The number 7^{1995} when divided by 100 leaves the remainder

- (A) 43 (B) 45 (C) 50 (D) none of these

Ans. A

Sol. $7^{1995} = 7[(7)^2]^{997} = 7(50 - 1)^{997}$
using expansion
 $7\{ {}^{997}C_0.50^{997} + {}^{997}C_1.50^{996} + \dots + {}^{997}C_{996}.50 - 1 \}$
 $= 100K + 7(997 \times 50 - 1)$
 $= 100K + 7(900 \times 50 + 7 \times 50 - 1)$
 $= 100K_1 + 49 \times 50 - 7 = 100K_1 + 2450 - 7$
 $= 100K_1 + 2400 + 43$
 $\therefore$ Remainder is 43

Q20. The last digit of $(3^P + 2)$, where $P = 3^{4n+2}$ is

- (A) 1 (B) 9 (C) 4 (D) 5

Ans. D

Sol. $(3^P + 2)$; where $P = 3^{4n+2}$

$$P = 3^{4n} \cdot 9 = 81^n \cdot 9$$

$$P = 9 \cdot (80 + 1)^n$$

$$= 9 \sum_{r=0}^n {}^nC_r (80)^{n-r}$$

$$P = 10\lambda + 9 \cdot {}^nC_n$$

$$\Rightarrow \text{Remainder} = 9 \text{ last digit of } (3^{3^{4n+2}} + 2) = \text{last digit of } 3^9 + 2$$

$$= 9^4 \cdot 3 + 2$$

$$= (10 - 1)^4 \cdot 3 + 2$$

$$= 3 + 2 + 10\lambda$$

$$\therefore \text{Remainder} = 5$$

Q21. The remainder left out when $8^{2n} - (62)^{2n+1}$ is divided by 9 is -

- (A) 0 (B) 2 (C) 7 (D) 8

Ans. B

Sol. -

Q22. If $X = \{4n - 3n - 1 : n \in \mathbb{N}\}$ and $Y = \{9(n - 1) : n \in \mathbb{N}\}$, where $\mathbb{N}$ is the set of natural numbers, then

$X \cup Y$ is equal to :

- (A) X (B) Y (C) N (D) $Y - X$

Ans. B

Sol. -

Q23. Only 4th term is numerically greatest term in $\left(2 + \frac{3x}{8}\right)^{10}$. Then the range of x is

- (A) $\left(\frac{-64}{21}, -2\right) \cup \left(1, \frac{64}{25}\right)$ (B) $\left(\frac{-64}{21}, -2\right) \cup \left(2, \frac{64}{21}\right)$ (C) $\left(-2, \frac{-64}{21}\right) \cup \left(2, \frac{64}{21}\right)$
 (D) $\left(\frac{-64}{21}, -2\right) \cup \left(1, \frac{64}{21}\right)$

Ans. B

Sol. We have the expansion of $\left(2 + \frac{3x}{8}\right)^{10}$

The general term is

$$T_{r+1} = \binom{10}{r} 2^{10-r} \left(\frac{3x}{8}\right)^r = \binom{10}{r} 3^r 2^{10-4r} x^r$$

We want the 4th term ($r = 3$) to be the numerically greatest term.

That means :

$$|T_4| \geq |T_3| \text{ and } |T_5| \leq |T_4|,$$

with strict inequalities for "only the 4th term."

Form $|T_4| \geq |T_3|$:

$$\frac{\binom{10}{3} 3^3 2^{-2} |x|^3}{\binom{10}{2} 3^2 2^2 |x|^2} \geq 1 \Rightarrow \frac{|x|}{2} \geq 1 \Rightarrow |x| \geq 2.$$

Equality at $|x| = 2$ gives $|T_4| = |T_3| \rightarrow$ excluded.

From $|T_5| \leq |T_4|$:

$$\frac{\binom{10}{3} 3^4 2^{-6} |x|^4}{\binom{10}{3} 3^3 2^{-2} |x|^3} \leq 1$$

$$\Rightarrow \frac{21}{64} |x| \leq 1 \Rightarrow |x| \leq \frac{64}{21}$$

Equality at $|x| = \frac{64}{21}$ gives $|T_5| = |T_4| \rightarrow$ excluded.

So strictly :

$$2 < |x| < \frac{64}{21}$$

Thus :

$$x \in \left(-\frac{64}{21}, -2\right) \cup \left(2, \frac{64}{21}\right)$$

Q24. If n is a positive integer, then $(\sqrt{3} + 1)^{2n} - (\sqrt{3} - 1)^{2n}$ is :

- (A) A rational number other than positive integers (B) An irrational number
(C) An odd positive integer (D) An even positive integer

Ans. B

Sol. -

Q25. $\sum_{r=1}^n r \cdot {}^{2n}C_r$ is equal to

- (A) $n \cdot 2^{n-1}$ (B) 2^{2n-1} (C) $2^{n-1} + 1$ (D) $n \cdot 2^{2n-1}$

Ans. D

Sol. -

Q26. If $C_0, C_1, C_2, \dots, C_n$ denote the coefficients in the expansion of $(1+x)^n$, then the value of $\sum_{r=0}^n (r+1)C_r$ is

- (A) $n \cdot 2^n$ (B) $(n+1)2^{n-1}$ (C) $(n+2)2^{n-1}$ (D) $(n+2)2^{n-2}$

Ans. C

Sol. -

Q27. If $(1+x)^n = C_0 + C_1 x + C_2 x^2 + \dots + C_n x^n$, then $C_0 + 5C_1 + 9C_2 + \dots + (4n+1)C_n$ is equal to

- (A) $n \cdot 2^n$ (B) $(n+1)2^n$ (C) $(2n+1)2^n$ (D) $(4n+1)2^n$

Ans. C

Sol. -

Q28. If $(1-x+x^2)^n = a_0 + a_1 x + a_2 x^2 + \dots + a_{2n} x^{2n}$, then $a_0 + a_2 + a_4 + \dots + a_{2n}$ equals

- (A) $\frac{3^n + 1}{2}$ (B) $\frac{3^n - 1}{2}$ (C) $\frac{1 - 3^n}{2}$ (D) $3^n + \frac{1}{2}$

Ans. A

Sol. -

Q29. If $(1+x)^n = C_0 + C_1 x + C_2 x^2 + \dots + C_n x^n$, then $\frac{C_1}{C_0} + \frac{2C_2}{C_1} + \frac{3C_3}{C_2} + \dots + \frac{nC_n}{C_{n-1}} =$

- (A) $\frac{n}{2}$ (B) $n(n+1)$ (C) $\frac{n(n+1)}{12}$ (D) $\frac{n(n+1)}{2}$

Ans. D

Sol. -

Q30. If $S_n = \sum_{r=0}^n \frac{1}{{}^nC_r}$ and $t_n = \sum_{r=0}^n \frac{r}{{}^nC_r}$, then $\frac{t_n}{s_n}$ is equal to-

- (A) $\frac{1}{2}n$ (B) $\frac{1}{2}n - 1$ (C) $n - 1$ (D) $\frac{2n-1}{2}$

Ans. A

Sol. $S_n = \sum_{r=0}^n \frac{1}{{}^nC_r}$

$$t_n = \sum_{r=0}^n \frac{r}{n C_r}$$

$$t_n = \sum_{r=0}^n \frac{n-r}{n C_r} \quad \therefore 2t_n = n \sum_{r=0}^n \frac{1}{n C_r} = n s_n \Rightarrow \frac{t_n}{s_n} = \frac{n}{2}$$

Q31. The sum of the series ${}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \dots - \dots + {}^{20}C_{10}$ is-

- (A) $-{}^{20}C_{10}$ (B) $\frac{1}{2} {}^{20}C_{10}$ (C) 0 (D) ${}^{20}C_{10}$

Ans. B

Sol. $s = {}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \dots + {}^{20}C_{10}$

$$s = {}^{20}C_{10} - {}^{20}C_{11} + \dots + {}^{20}C_{20}$$

$$2s = 0 + {}^{20}C_{10}$$

$$s = \frac{1}{2} ({}^{20}C_{10})$$

Q32. The sum of coefficients of integral powers of x in the binomial expansion of $(1 - 2\sqrt{x})^{50}$ is

- (A) $\frac{1}{2}(3^{50}+1)$ (B) $\frac{1}{2}(3^{50})$ (C) $\frac{1}{2}(3^{50}-1)$ (D) $\frac{1}{2}(2^{50}+1)$

Ans. A

Sol. $(1 - 2\sqrt{x})^{50} = 1 - {}^{50}C_1(2\sqrt{x}) + {}^{50}C_2(x^2)(\sqrt{x})^2 + \dots$

Put $\sqrt{x}=1$ & $\sqrt{x}=-1$ & then add

Ans, $\frac{3^{50}+1}{2}$

Q33. The value of $({}^{21}C_1 - {}^{10}C_1) + ({}^{21}C_2 - {}^{10}C_2) + ({}^{21}C_3 - {}^{10}C_3) + ({}^{21}C_4 - {}^{10}C_4) + \dots + ({}^{21}C_{10} - {}^{10}C_{10})$ is

- (A) $2^{20} - 2^9$ (B) $2^{20} - 2^{10}$ (C) $2^{21} - 2^{11}$ (D) $2^{21} - 2^{10}$

Ans. B

Sol. -

Q34. The sum of the co-efficients of all odd degree terms in the expansion of $(x + \sqrt{x^3 - 1})^5 + (x - \sqrt{x^3 - 1})^5$, ($x > 1$) is

- (A) 1 (B) 2 (C) -1 (D) 0

Ans. B

Sol. $f(x) = (x + \sqrt{x^3 - 1})^5 + (x - \sqrt{x^3 - 1})^5$

$$= 2(x^5 + 10x^3(x^3 - 1) + 5x(x^3 - 1)^2)$$

$$\text{Sum of coefficients of odd degree} = \frac{f(1) - f(-1)}{2} = 2$$

- Q35. For $r = 0, 1, \dots, 10$, let A_r , B_r and C_r denote, respectively, the coefficient of x^r in the expansions of $(1+x)^{10}$, $(1+x)^{20}$ and $(1+x)^{30}$. Then $\sum_{r=1}^{10} A_r (B_{10} B_r - C_{10} A_r)$ is equal to
- (A) $B_{10} - C_{10}$ (B) $A_{10} (B_{10}^2 - C_{10} A_{10})$ (C) 0 (D) $C_{10} - B_{10}$

Ans. D

Sol. $A_r = {}^{10}C_r$, $B_r = {}^{20}C_r$, $C_r = {}^{30}C_r$

$$\begin{aligned} & \sum_{r=1}^{10} \left({}^{20}C_{10} {}^{10}C_r {}^{20}C_r - {}^{30}C_{10} \left({}^{10}C_r \right)^2 \right) \\ &= {}^{20}C_{10} \left({}^{10}C_1 {}^{20}C_1 + {}^{10}C_2 {}^{20}C_2 + \dots + {}^{10}C_{10} {}^{20}C_{10} \right) \\ & \quad - {}^{30}C_{10} \left({}^{10}C_1^2 + {}^{10}C_2^2 + \dots + {}^{10}C_{10}^2 \right) \\ &= {}^{20}C_{10} \left({}^{30}C_{10} - 1 \right) - {}^{30}C_{10} \left({}^{20}C_{10} - 1 \right) \\ &= {}^{30}C_{10} - {}^{20}C_{10} = C_{10} - B_{10} \end{aligned}$$

- Q36. Number of terms in the expansion of $(x^{1/3} + x^{2/5})^{40}$ with integral power of x is equal to

Ans. 3

Sol. $T_{r+1} = {}^{40}C_r x^{\frac{40-r}{3}} \cdot x^{\frac{2r}{5}}$
 $= {}^{40}C_r x^{\frac{200+r}{15}}$

$r = 10, 25, 40$

$\Rightarrow$ Number of such terms = 3

Exercise J02

- Q1. (a) If the coefficients of $(2r + 4)^{\text{th}}$, $(r - 2)^{\text{th}}$ terms in the expansion of $(1 + x)^{18}$ are equal, find r .
 (b) If the coefficients of the r^{th} , $(r+1)^{\text{th}}$ & $(r+2)^{\text{th}}$ terms in the expansion of $(1+x)^{14}$ are in AP, find r .
 (c) If the coefficients of 2^{nd} , 3^{rd} & 4^{th} terms in the expansion of $(1+x)^{2n}$ are in AP, show that

$$2n^2 - 9n + 7 = 0.$$

Ans. (a) $r = 6$ (b) $r = 5$ or 9

Sol. -

- Q2. (a) Find the term independent of x in the expansion of (i) $\left[\sqrt{\frac{x}{3}} + \frac{\sqrt{3}}{2x^2}\right]^{10}$ (ii) $\left[\frac{1}{2}x^{1/3} + x^{-1/5}\right]^8$
 (b) Find the value of x for which the fourth term in the expansion, $\left(5^{\frac{2}{5}\log_5 \sqrt{4x+44}} + \frac{1}{5^{\log_5 \sqrt[3]{2^{x-1}+7}}}\right)^8$ is 336.

Ans. (a) (i) $\frac{5}{12}$ (ii) $T_6 = 7$, (b) $x = 0$ or 1

Sol. -

- Q3. Prove that the ratio of the coefficient of x^{10} in $(1 - x^2)^{10}$ & the term independent of x in $\left(x - \frac{2}{x}\right)^{10}$ is $1 : 32$.

Ans. -

Sol. -

- Q4. Let a and b be the coefficient of x^3 in $(1 + x + 2x^2 + 3x^3)^4$ and $(1 + x + 2x^2 + 3x^3 + 4x^4)^4$ respectively. Find the value of $(a - b)$.

Ans. 0

Sol. -

- Q5. Find the coefficient of xr in the expression of $(x + 3)^{n-1} + (x + 3)^{n-2}(x + 2) + (x + 3)^{n-3}(x + 2)^2 + \dots + (x + 2)^{n-1}$.

Ans. ${}^nC_r (3^{n-r} - 2^{n-r})$

Sol. -

- Q6. Find the term independent of x in the expansion of $(1 + x + 2x^3) \left(\frac{3x^2}{2} - \frac{1}{3x}\right)^9$.

Ans. $\frac{17}{54}$

Sol. -

Q7. Let $(1+x^2)^2 \cdot (1+x)^n = \sum_{k=0}^{n+4} a_k \cdot x^k$. If a_1, a_2 & a_3 are in AP, find n .

Ans. $n = 2$ or 3 or 4

Sol. -

Q8. Find the sum of the roots (real or complex) of the equation $x^{2001} + \left(\frac{1}{2} - x\right)^{2001} = 0$.

Ans. 500

Sol. -

Q9. Find numerically greatest term in the expansion of
(i) $(2 + 3x)^9$ when $x = \frac{3}{2}$ (ii) $(3 - 5x)^{15}$ when $x = \frac{1}{5}$

Ans. (i) $T_7 = \frac{7 \cdot 3^{13}}{2}$ (ii) 455×3^{12}

Sol. -

Q10. Find the index n of the binomial $\left(\frac{x}{5} + \frac{2}{5}\right)^n$ if the 9th term of the expansion has numerically the greatest coefficient ($n \in \mathbb{N}$).

Ans. $n = 12$

Sol. -

Q11. For which positive values of x is the fourth term in the expansion of $(5 + 3x)^{10}$ is the greatest.

Ans. $\frac{5}{8} < x < \frac{20}{21}$

Sol. -

Q12. Find the coefficient of
(a) $x^2 y^3 z^4$ in the expansion of $(ax - by + cz)^9$.
(b) $a^2 b^3 c^4 d$ in the expansion of $(a - b - c + d)^{10}$.

Ans. (a) $-1260 \cdot a^2 b^3 c^4$; (b) -12600

Sol. -

Q13. Find the coefficient of
(a) x^4 in the expansion of $(1 + x + x^2 + x^3)^{11}$ (b) x^4 in the expansion of $(2 - x + 3x^2)^6$.

Ans. (a) 990 (b) 3660

Sol. -

Q14. (a) Show that the integral part in each of the following is odd. $n \in \mathbb{N}$

(A) $(5 + 2\sqrt{6})^n$ (B) $(8 + 3\sqrt{7})^n$

(b) Show that the integral part in each of the following is even. $n \in \mathbb{N}$

(A) $(3\sqrt{3} + 5)^{2n+1}$ (B) $(5\sqrt{5} + 11)^{2n+1}$

Ans. -

Sol. -

Q15. If $(7 + 4\sqrt{3})^n = p + \beta$ where n & p are positive integers and β is a proper fraction show that

$$(1 - \beta)(p + \beta) = 1.$$

Ans. -

Sol. -

Q16. If $(6\sqrt{6} + 14)^{2n+1} = N + F$ where N & F be the fractional part of N , prove that $NF = 202^{n+1}$ ($n \in \mathbb{N}$)

Ans. -

Sol. -

Q17. Let I denotes the integral part and F the proper fractional part of $(3 + \sqrt{5})^n$ where $n \in \mathbb{N}$ and if r denotes the rational part and σ the irrational part of the same, show that $\rho = \frac{1}{2}(I + 1)$ and $\sigma = \frac{1}{2}(I + 2F - 1)$

Ans. -

Sol. -

Q18. Given $s_n = 1 + q + q^2 + \dots + q^n$ & $S_n = 1 + \frac{q+1}{2} + \left(\frac{q+1}{2}\right)^2 + \dots + \left(\frac{q+1}{2}\right)^n$, $q \neq 1$, prove that ${}^{n+1}C_1 + {}^{n+1}C_2 \cdot s_1 + {}^{n+1}C_3 \cdot s_2 + \dots + {}^{n+1}C_{n+1} \cdot s_n = 2^n \cdot S_n$.

Ans. -

Sol. -

Q19. Find the sum of the series $\sum_{r=0}^n (-1)^r \cdot {}^nC_r \left[\frac{1}{2^r} + \frac{3^r}{2^{2r}} + \frac{7^r}{2^{3r}} + \frac{15^r}{2^{4r}} + \dots \right]$ up to m terms

Ans. $\frac{(2^{mn} - 1)}{(2^n - 1)(2^{mn})}$

Sol. -

Q20. Using the theory of permutation or otherwise prove that

$$^{100}C_{10} + 5 \cdot ^{100}C_{11} + 10 \cdot ^{100}C_{12} + 10 \cdot ^{100}C_{13} + 5 \cdot ^{100}C_{14} + ^{100}C_{15} = ^{105}C_{90}$$

Ans. -

Sol. -

Q21. Let $f(n) = \sum_{r=0}^n \sum_{k=r}^n \binom{k}{r}$. Find the total number of divisors of $f(11)$.

Ans. 24

Sol. -

Q22. Given that $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n}$, find the values of :

(i) $a_0 + a_1 + a_2 + \dots + a_{2n}$; (ii) $a_0 - a_1 + a_2 - a_3 + \dots + a_{2n}$; (iii) $a_0^2 - a_1^2 + a_2^2 - a_3^2 + \dots + a_{2n}^2$

Ans. (i) 3^n (ii) 1, (iii) a_n

Sol. -

Q23. Prove that $\sum_{k=0}^n {}^nC_k \sin Kx \cdot \cos(n - K)x = 2^{n-1} \sin nx$.

Ans. -

Sol. -

Q24. Find the number of divisors of the number

$$N = {}^{2000}C_1 + 2 \cdot {}^{2000}C_2 + 3 \cdot {}^{2000}C_3 + \dots + 2000 \cdot {}^{2000}C_{2000}.$$

Ans. 8016

Sol. -

Q25. Prove the following identities using the theory of permutation where $C_0, C_1, C_2, \dots, C_n$ are the combinatorial coefficients in the expansion of $(1 + x)^n$, $n \in \mathbb{N}$, then prove the following:

(a) $C_0^2 + C_1^2 + C_2^2 + \dots + C_n^2 = \frac{(2n)}{n!n!}$

(b) $C_0 C_1 + C_1 C_2 + C_2 C_3 + \dots + C_{n-1} C_n = \frac{(2n)!}{(n+1)!(n-1)!}$

(c) $C_0 C_r + C_1 C_{r+1} + C_2 C_{r+2} + \dots + C_{n-r} C_n = \frac{(2n)!}{(n-r)!(n+r)!}$

(d) $\sum_{r=0}^{n-2} \left({}^nC_r \cdot {}^nC_{r+2} \right) = \frac{(2n)!}{(n-2)!(n+2)!}$

Ans. -

Sol. -

Q26. Prove the following identities using the theory of permutation where $C_0, C_1, C_2, \dots, C_n$ are the combinatorial coefficients in the expansion of $(1+x)^n$, $n \in \mathbb{N}$, then prove the following:

(a) $C_1 + 2C_2 + 3C_3 + \dots + n \cdot C_n = n \cdot 2^{n-1}$

(b) $C_0 + 2C_1 + 3C_2 + \dots + (n+1)C_n = (n+2)2^{n-1}$

(c) $C_0 + 3C_1 + 5C_2 + \dots + (2n+1)C_n = (n+1)2^n$

(d) $(C_0+C_1)(C_1+C_2)(C_2+C_3) \dots (C_{n-1}+C_n) = \frac{C_0 \cdot C_1 \cdot C_2 \cdot \dots \cdot C_{n-1} (n+1)^n}{n!}$

(e) $1 \cdot C_0^2 + 3 \cdot C_1^2 + 5 \cdot C_2^2 + \dots + (2n+1) C_n^2 = \frac{(n+1)(2n)!}{n!n!}$

Ans. -

Sol. -

Paragraph

A path of length n is a sequence of points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ with integer coordinates

such that for all i between 1 and $n-1$ both inclusive,

neither $x_{i+1} = x_i + 1$ and $y_{i+1} = y_i$ (in which case we say the i th step is rightward)

nor $x_{i+1} = x_i$ and $y_{i+1} = y_i + 1$ (in which case we say that the i th step is upward).

This path is said to start at (x_1, y_1) and end at (x_n, y_n) . Let $P(a, b)$, for a and b non-negative integers, denotes the number of paths that start at $(0, 0)$ and end at (a, b) .

Q27. The value of $\sum_{i=0}^{10} P(i, 10-i)$, is

- (A) 1024 (B) 512 (C) 256 (D) 128

Ans. A

Sol. -

Paragraph

A path of length n is a sequence of points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ with integer coordinates

such that for all i between 1 and $n-1$ both inclusive,

neither $x_{i+1} = x_i + 1$ and $y_{i+1} = y_i$ (in which case we say the i th step is rightward)

nor $x_{i+1} = x_i$ and $y_{i+1} = y_i + 1$ (in which case we say that the i th step is upward).

This path is said to start at (x_1, y_1) and end at (x_n, y_n) . Let $P(a, b)$, for a and b non-negative integers, denotes the number of paths that start at $(0, 0)$ and end at (a, b) .

Q28. Number of ordered pairs (i, j) where $i \neq j$ for which $P(i, 100-i) = P(j, 100-j)$, is

- (A) 50 (B) 99 (C) 100 (D) 101

Ans. C

Sol. -

Paragraph

A path of length n is a sequence of points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ with integer coordinates

such that for all i between 1 and $n - 1$ both inclusive,

neither $x_{i+1} = x_i + 1$ and $y_{i+1} = y_i$ (in which case we say the i th step is rightward)

nor $x_{i+1} = x_i$ and $y_{i+1} = y_i + 1$ (in which case we say that the i th step is upward).

This path is said to start at (x_1, y_1) and end at (x_n, y_n) . Let $P(a, b)$, for a and b non-negative integers, denotes the number of paths that start at $(0, 0)$ and end at (a, b) .

Q29. The sum $P(43, 4) + \sum_{j=1}^5 (49 - j, 3)$ is equal to

- (A) $P(4, 48)$ (B) $P(3, 49)$ (C) $P(4, 47)$ (D) $P(5, 47)$

Ans. A

Sol. -

Paragraph

Let $S_k = 3^k \binom{100}{k} \binom{100}{n-k} - 3^{k-2} \binom{100}{k-2} \binom{98}{n-k-2} - \dots + (-1)^k \binom{100}{k} \binom{100-k}{n}$

where $\binom{n}{r}$ represents the coefficient of x^r in the binomial expansion of $(1 + x)^n$

$\forall n \in \mathbb{N}$

and $V_k = \left(\frac{1}{2}\right)^k S_k = M(100, k)$.

Q30. The value of $\sum_{k=0}^{100} S_k S_{100-k}$ equals

- (A) $\binom{200}{100}$ (B) $2^{100} \binom{200}{100}$ (C) $2^{200} \binom{200}{100}$ (D) $2^{99} \binom{200}{99}$

Ans. B

Sol. -

Paragraph

Let $S_k = 3^k \binom{100}{0} \binom{100}{k} - 3^{k-2} \binom{100}{2} \binom{98}{k-2} - \dots + (-1)^k \binom{100}{k} \binom{100-k}{0}$
 where $\binom{n}{r}$ represents the coefficient of x^r in the binomial expansion of $(1+x)^n$

$\forall n \in \mathbb{N}$

and $V_k = \left(\frac{1}{2}\right)^k S_k = M(100, k)$.

Q31. The value of the $M(100, 49) + M(100, 50)$ is equal to

- (A) $M(99, 50)$ (B) $M(100, 51)$ (C) $M(101, 49)$ (D) $M(101, 51)$

Ans. D

Sol. -

Paragraph

Let $S_k = 3^k \binom{100}{0} \binom{100}{k} - 3^{k-2} \binom{100}{2} \binom{98}{k-2} - \dots + (-1)^k \binom{100}{k} \binom{100-k}{0}$
 where $\binom{n}{r}$ represents the coefficient of x^r in the binomial expansion of $(1+x)^n$

$\forall n \in \mathbb{N}$

and $V_k = \left(\frac{1}{2}\right)^k S_k = M(100, k)$.

Q32. In an examination, there are 100 true-false type questions. The number of ways in which one student can make his choice to attempt or not to attempt the questions is given by

- (A) $\sum_{k=0}^{100} S_k$ (B) $\sum_{k=0}^{100} V_k$ (C) $\sum_{k=1}^{100} S_k$ (D) $\sum_{k=1}^{100} V_k$

Ans. A

Sol. -

Q33. In the expansion of $\left(x - \frac{\alpha}{x}\right)^n$ and $\left(x + \frac{\beta}{x^2}\right)^n$ in powers of x have one term independent of x , then n is divisible by

- (A) 2 (B) 3 (C) 4 (D) 6

Ans. D, B, A

Sol. -

Q34. In the expansion of $(a + b + c)^{10}$

- (A) total number of terms is 66 (B) coefficient of $a^8 bc$ is 90
 (C) coefficient of $a^4 b^5 c^3$ is 0 (D) none of these

Ans. C, B, A

Sol. -

Q35. In the expansion of $\left(x + \frac{\alpha}{x^2}\right)^n$, $\alpha \neq 0$, if no term is independent of x , then n is

- (A) 10 (B) 12 (C) 16 (D) 20

Ans. A, C, D

Sol. -

Q36. In the expansion of $\left(x + \frac{1}{x} + 2\right)^{2009}$

- (A) Middle term is $^{4018}C_{2009}$ (B) The term independent of x is $^{4018}C_{2009}$
 (C) Coefficient of x in the expansion is $^{4018}C_{2010}$
 (D) Coefficient of $1/x$ in the expansion is $^{4018}C_{2008}$

Ans. B, A, C, D

Sol. -

Q37. Which of the following statement(s) is/are correct?

- (A) $1 + \frac{2}{2} + \frac{3}{2^2} + \frac{4}{2^3} + \dots + \infty = 4$
 (B) Integral part of $(9 + 4\sqrt{5})^n$, is even, $n \in \mathbb{N}$.
 (C) $({}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n)^2 = 1 + {}^{2n}C_1 + {}^{2n}C_2 + \dots + {}^{2n}C_{2n}$
 (D) $\frac{1}{(3+2x)^2}$ can be expanded as infinite series in ascending powers of x only if $|x| < \frac{2}{3}$.

Ans. C, A

Sol. -

Q38. If $\left(9 + \sqrt{80}\right)^n = 1 + f$ where I, n are integers and $0 < f < 1$, then

- (A) I is an odd integer (B) I is an even integer (C) $(I + f)(1 - f) = 1$
 (D) $1 - f = (9 - \sqrt{80})^n$

Ans. D, C, A

Sol. -

Q39. Let $(1 + x^2)^2 (1 + x)^n = A_0 + A_1 x + A_2 x^2 + \dots$. If A_0, A_1, A_2 are in A.P. then the value of n is

- (A) 2 (B) 3 (C) 5 (D) 7

Ans. A, B

Sol. -

Q40. If ${}^{100}C_6 + 4 \cdot {}^{100}C_7 + 6 \cdot {}^{100}C_8 + 4 \cdot {}^{100}C_9 + {}^{100}C_{10}$ has the value equal to ${}^x C_y$; then the possible value(s) of $x + y$ can be :

- (A) 112 (B) 114 (C) 196 (D) 198

Ans. D, B

Sol. -

Q41. **Match the column**

Column I

Column II

(A) If the coefficients of x^7 and x^8 in $\left(2 + \frac{x}{3}\right)^n$ are equal, then n is (P) 2

(B) If the coefficient of x in the expansion of $\left(x - \frac{1}{ax^2}\right)^{10}$ is -15 ,
 then the value of a , is (Q)

01

(C) The remainder when 2^{2003} is divided by 17 is (R)
 55

(D) The last two digits of 3^{400} , are (S)
 8

(A) (A) – R (B) – P (C) – S (D) – Q (B) (A) – S (B) – P (C) – R (D) – Q

(C) (A) – R (B) – Q (C) – S (D) – P (D) (A) – Q (B) – P (C) – S (D) – R

Ans. 5eed25c104b8298d728a1b2

Sol. -